Probabilistic School Bus Routing with Stop Selection

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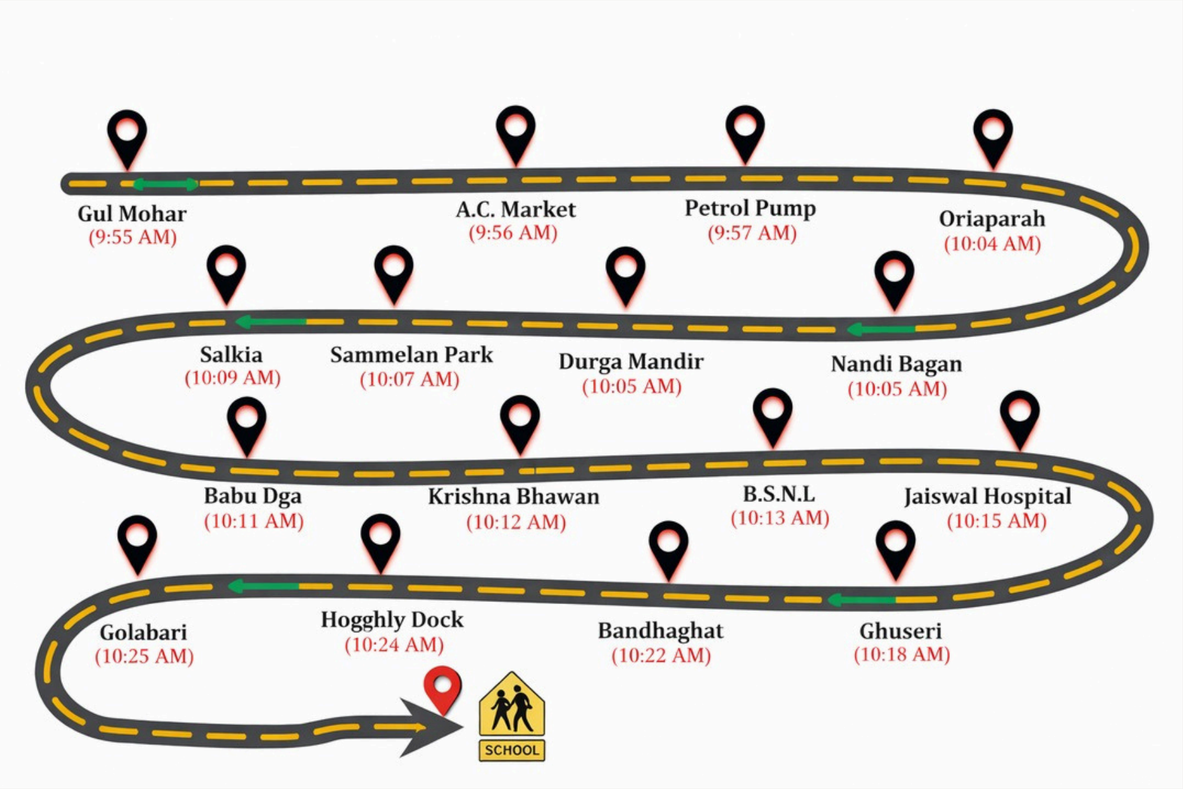


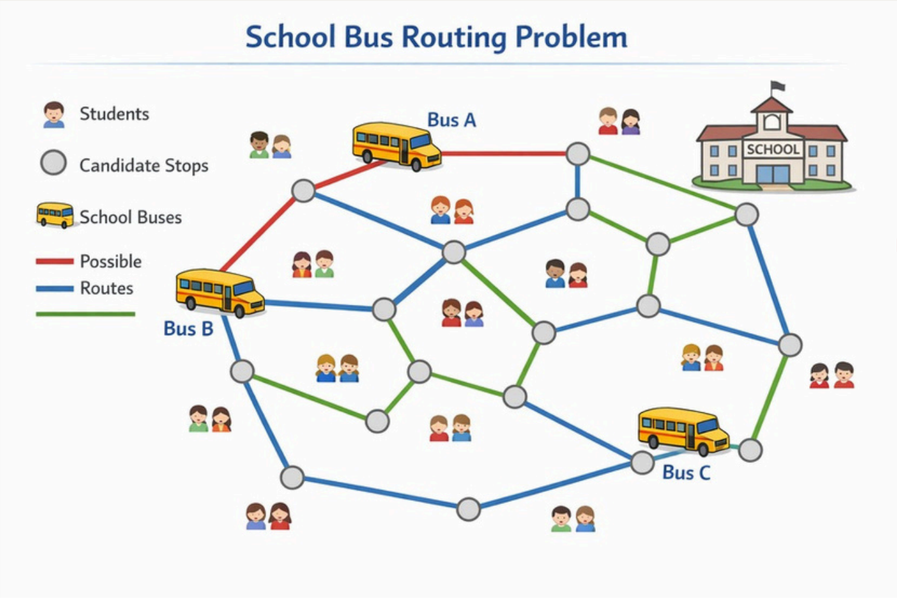
Made by: Virja Shah (AU2440036) & Aryan Thakkar (AU2440248)

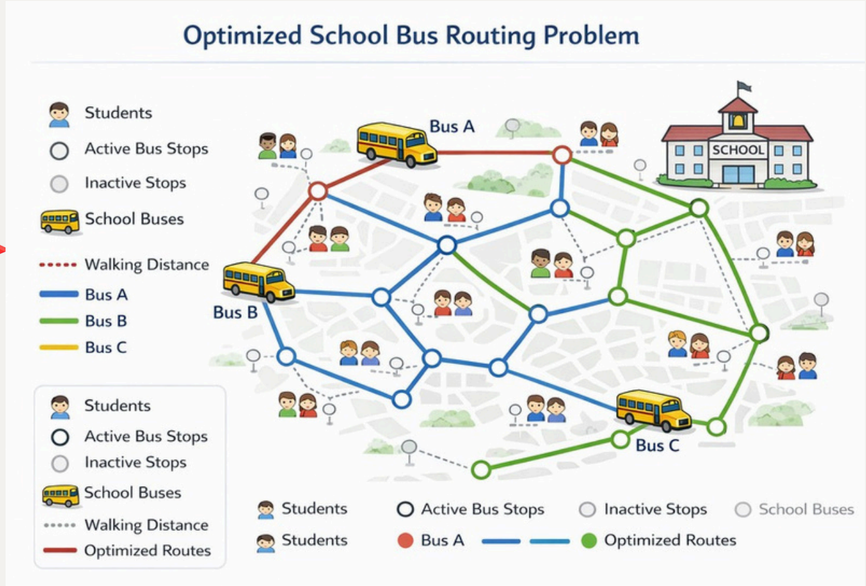
Project Context and Motivation

- **ApplicationDomain:**IntelligentTransportationSystem(ITS) [school bus - routing]

- **Real world context:** Student assignment to stops, route sequencing and traffic variability

- **Reason for uncertainty:** random daily demand, spatial distribution and traffic speed fluctuations





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Project Objective and Scope

Exact Objective:

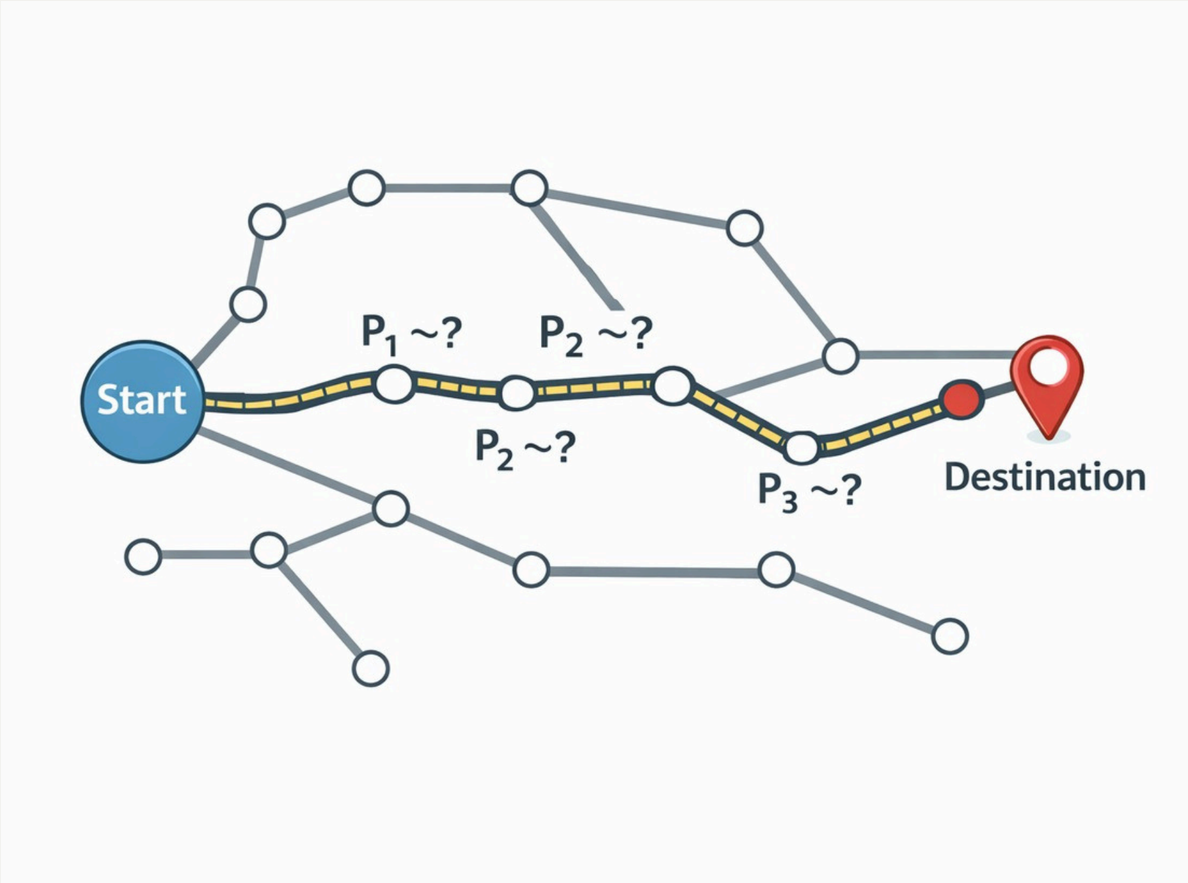
- Minimize expected travel time ofall school buses, minE[Troute]

Decision Task:

- Select stops, asign students and sequence routes

Scope:

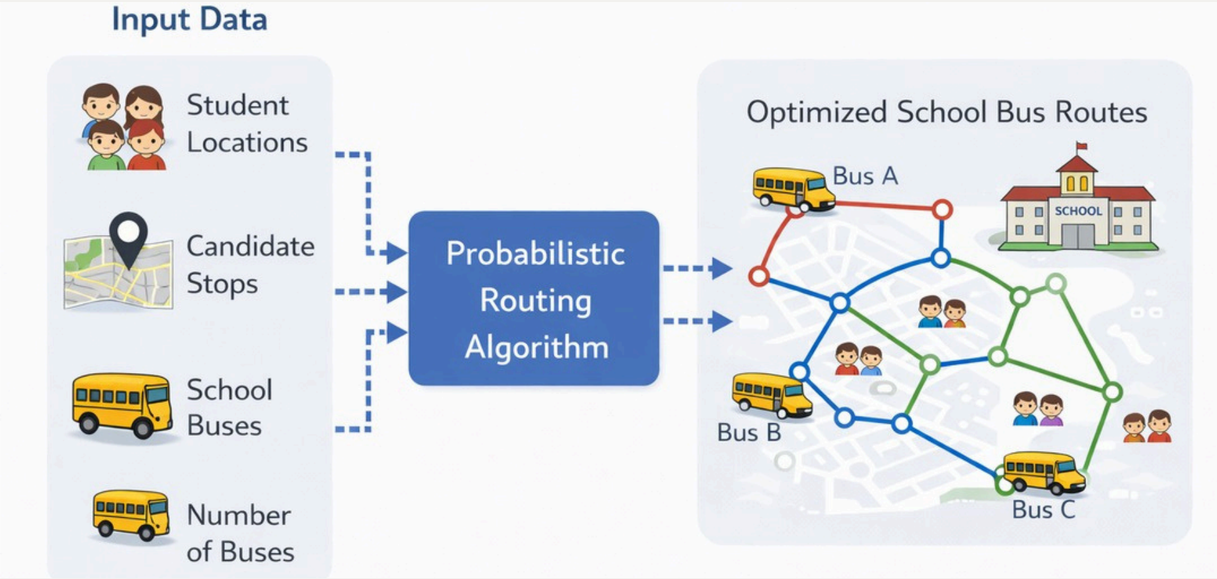
- Static road network, stochastic demand and traffic



System Pipeline

Generate student stop location → ***Compute distance matrix*** → ***Student demand at stops*** → ***Route construction using deterministic*** → ***Bus route generation*** → ***Travel distace computation*** → ***Performance evaluation***

- **Components:** Stop location, distance matrix, student stop, deterministic routing



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Source of Uncertainty in the Project

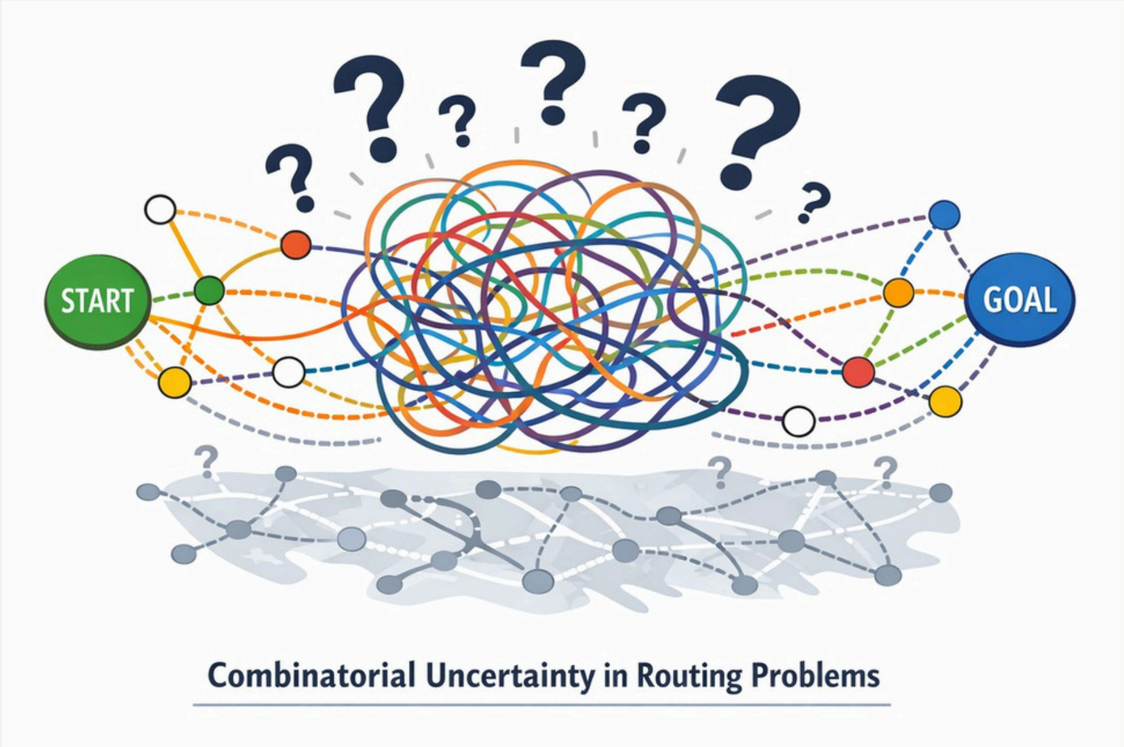
The routing algorithm is deterministic thus it may contain uncertainty.

Possible sources:

- Student locations variablity
- Changing of transportation demand
- Operational constraints

In simulation:

- Generations of input data to represent realistic routing.
- Algorithms operating on these inputs to give ouput as routes.



Key Random Variables and Their Roles

1. Stop locations - Each stop is represented by 2D coordinates
2. Depot locations - Starting and ending location of all buses
3. Distance Matrix dij - Distance between two stops i and j
4. Demand values - No of students at each stop

Probabilistic Modeling with deterministic decisions

Mathematical assumption-

$$X_i = (x_i, y_i)$$

Distances between stops are computed using the distance formula-

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

And we take the school as **node 0**

Uncertainty with Deterministic Rules

- The routing algorithm follows fixed decision rules.
- For the same dataset in input, the output will always be the same routing solution
- Randomness exists in inputs, and decision taken regarding routing are deterministic

Relevant Clarke-Wright Equation

$$S_{ij} = d_{i0} + d_{0j} - d_{ij}$$

Deterministic Baseline logic

Nearest Neighbour Heuristic

Steps:

1. Start at depot.
2. Select the closest stop that is unvisited
3. Add the stop to route
4. Repeat until all stops visited
5. Return to the depot

Clarke Wright savings Algorithm

Steps:

1. Assign each route to each stop
2. Calculate the distance saving

$$S_{ij} = d_{i0} + d_{0j} - d_{ij}$$

3. Sort savings values
4. Merge routes with highest savings
5. Continue until no margin remain

Simulation Implementation

Algorithms used:

- Nearest Neighbour Heuristic
- Clarke-Wright Savings Algorithm

Code Structure overview:

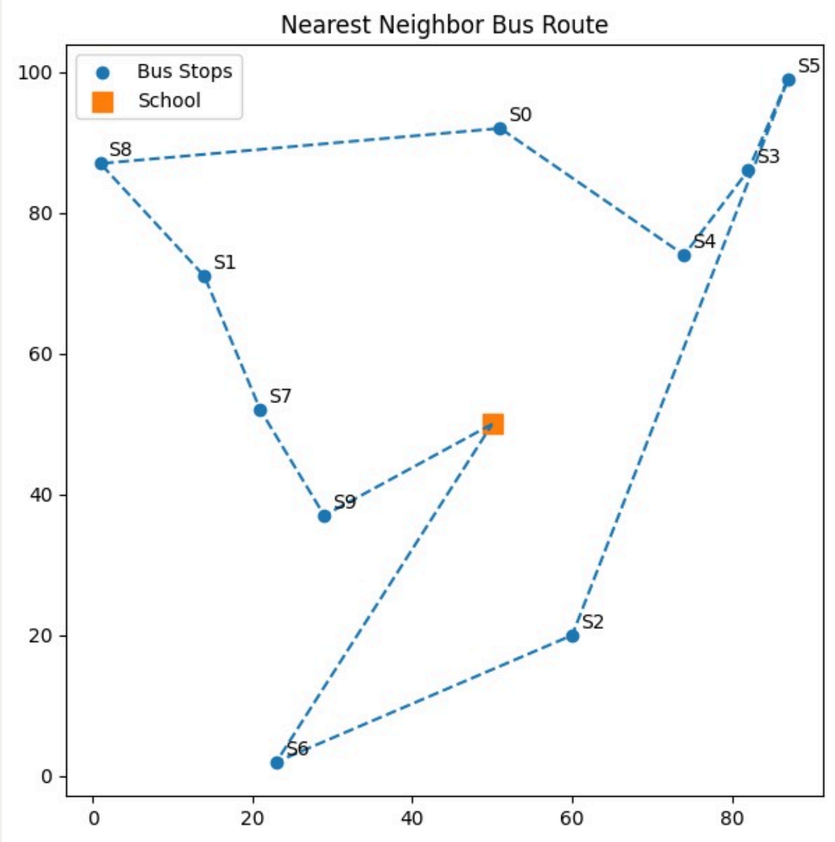
- Input Generation module → Distance matrix module → Routing Algorithms module → Route Evaluation Module

Full pipeline execution:

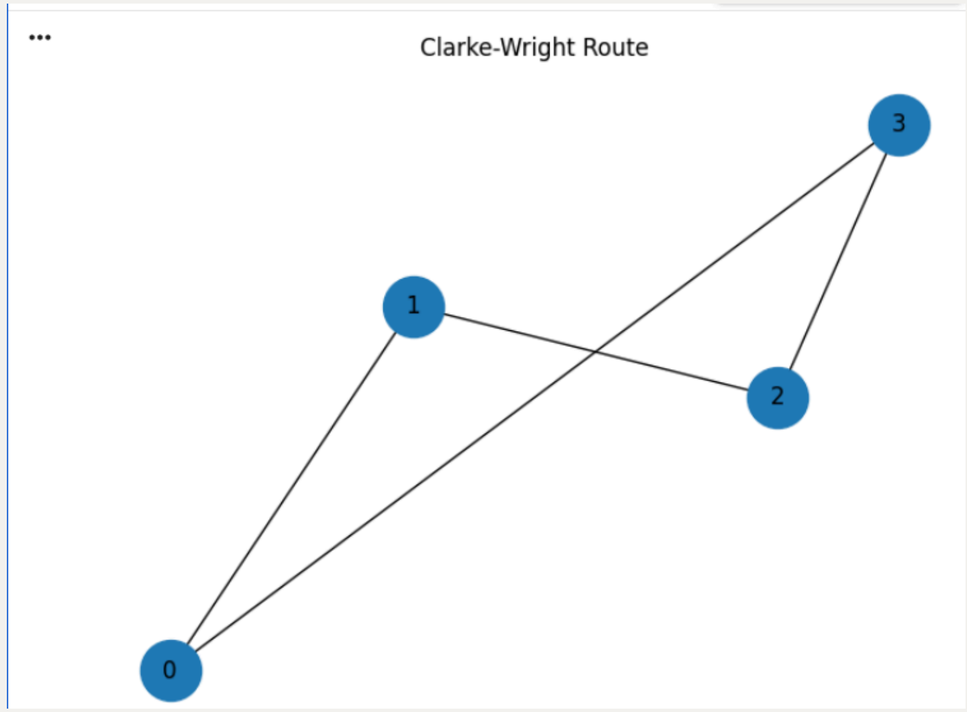
- Generate stop locations → calculate distance → routing algo → constructing route → calculate travel distance

Results and Performance analysis

Nearest Neighbour Heuristic



Clarke Wright savings Algorithm



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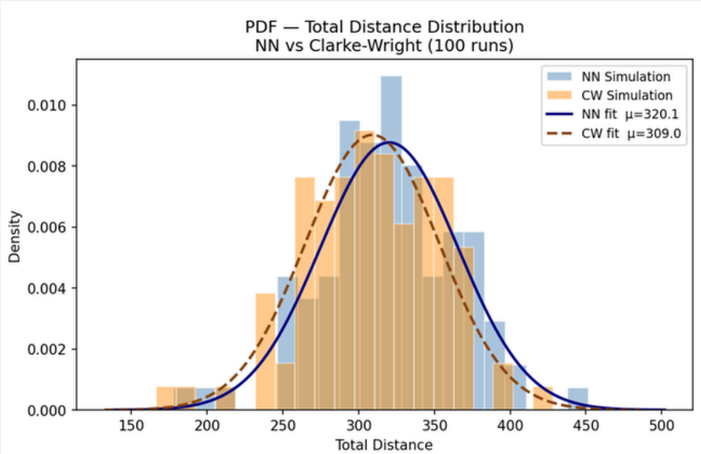

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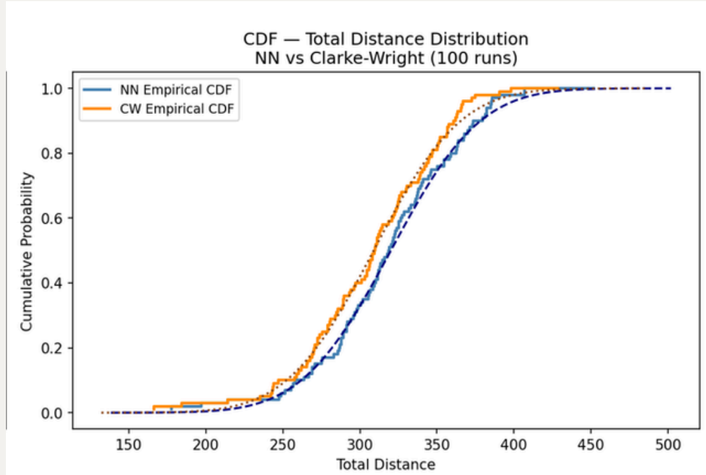

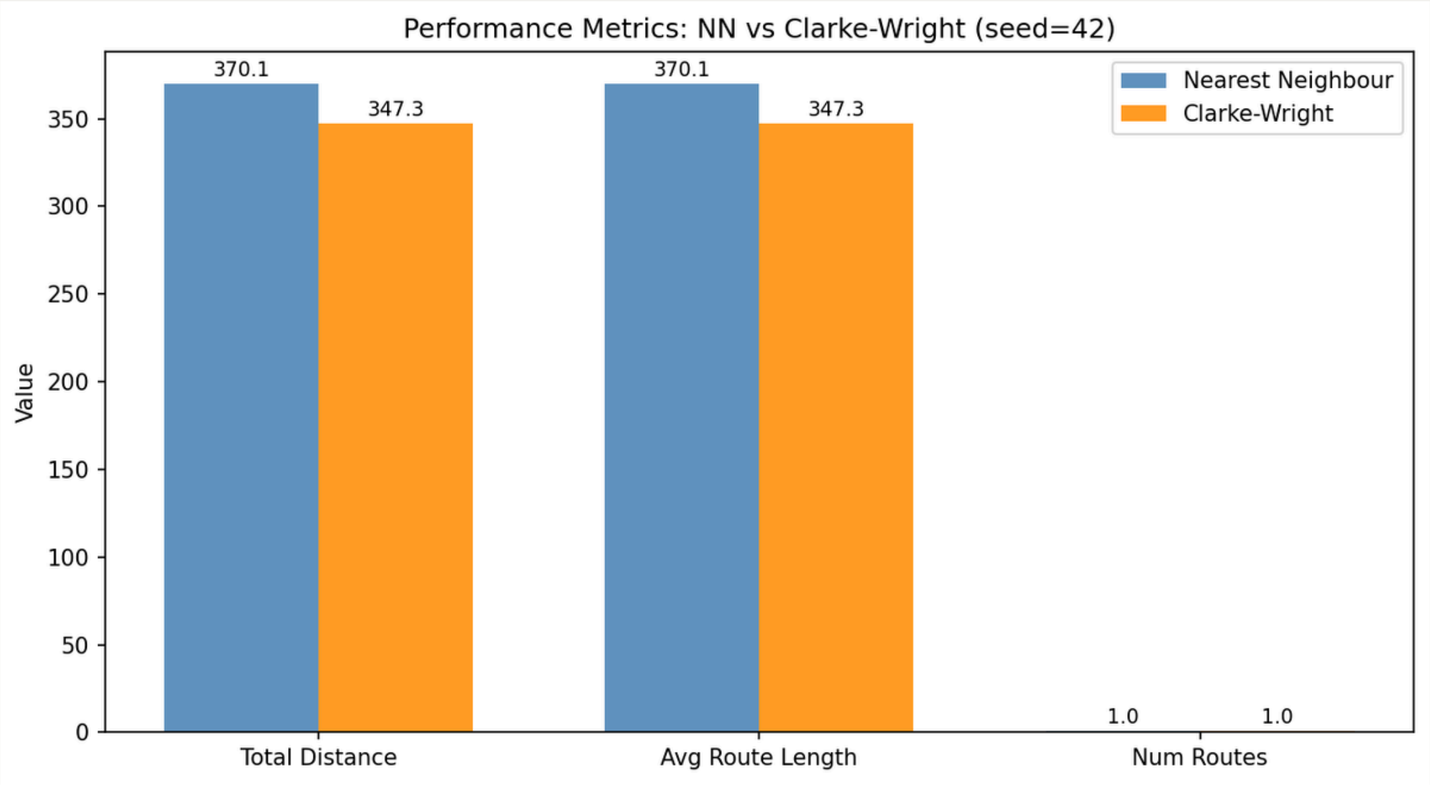

Comparision between the algorithm

Limitations of Deterministic Strategies

- No flexibility

The deterministic algorithms follow fixed rules and produce same results for same inputs at every run.

- Greedy Behaviour

Heuristics used such as Nearest Neighbour make optimal decisions, which give suboptimal routes globally

- Input data Sensitivity

Changes no matter how small in stop locations or in student demand, will affect the generated routes.

- Scalability issue

If the number of stops increase, deterministic heuristics used here might not give efficient routing solutions over time.

Summary and Future Direction

- Deterministic heuristics, i.e., Nearest Neighbour and Clarke-Wright algorithm give an efficient and basic solution to routing problems.
- The methods we used here, help in building feasible routes as well as to reduce overall travel distance and the computational complexity.
- But deterministic algorithms may seem to struggle in handling uncertainty and dynamic changes over time in real world transportation scenarios.
- Future work will focus include probabilistic models and stochastic simulations to better understand system variability.
- Thus, this approach will provide more robust and adaptive routing strategies for the School Bus Routing Problem (SBRP).

THANK YOU